

# Severe Congenital Neutropenia 7, Autosomal Recessive (SCN7) — Comprehensive Disease Report

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**Disease:** Severe Congenital Neutropenia 7, Autosomal Recessive **MONDO ID:** MONDO:0014865  
· **OMIM phenotype:** #616022 · **Causal gene:** *JAGN1* (OMIM 616012, HGNC:26926) *Category:*\*  
Mendelian (monogenic, autosomal recessive)

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## Summary

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Severe Congenital Neutropenia 7 (SCN7) is a rare autosomal recessive primary immunodeficiency / bone-marrow-failure disorder caused by **biallelic loss-of-function variants in *JAGN1*** (Jagunal homolog 1), a small, evolutionarily conserved endoplasmic reticulum (ER)-resident integral membrane protein. Whole-exome sequencing first defined the disorder in 2014, identifying 9 distinct homozygous *JAGN1* mutations in 14 affected individuals ([PMID: 25129144](#)). *JAGN1* deficiency is now estimated to account for **roughly 10% of all severe congenital neutropenia (SCN) cases** ([PMID: 41751872](#)).

Mechanistically, SCN7 is best understood as a **myeloid ER-secretory / glycosylation and G-CSF-receptor-signaling disorder**. Loss of *JAGN1* disrupts ER vesicular trafficking, protein N-glycosylation and sialylation, cytotoxic granule formation, and granulocyte colony-stimulating factor receptor (G-CSFR) signaling. This produces a **two-branch phenotype**: a *quantitative* defect (maturation arrest and calpain-dependent apoptosis of myeloid progenitors → severe neutropenia) and a *qualitative* defect (impaired neutrophil adhesion, migration, degranulation and microbial killing — present in mouse models even without frank neutropenia). Clinically, patients present in the neonatal period or early infancy with recurrent bacterial and fungal infections, aphthous stomatitis, and skin abscesses; about a third have a syndromic spectrum including short stature, facial dysmorphism, and bone/neurodevelopmental abnormalities.

Prognostically, SCN7 is a lifelong condition dominated by infection- and leukemia-related morbidity. Unlike *ELANE*- or *HAX1*-related SCN, **most *JAGN1* patients respond poorly to recombinant G-CSF**, and GM-CSF has also failed to rescue neutrophil counts in reported cases.

Like all SCN, it carries a premalignant risk of myelodysplastic syndrome / acute myeloid leukemia (MDS/AML). **Allogeneic hematopoietic stem cell transplantation (alloHSCT) is the only definitive, curative therapy** and is preferentially applied to poor G-CSF responders — the group that includes most *JAGN1*-deficient patients.

## Key Findings

### Finding 1 — SCN7 is caused by biallelic (autosomal recessive) loss-of-function variants in *JAGN1*

The genetic basis of SCN7 was established by whole-exome sequencing, which identified **9 distinct homozygous *JAGN1* mutations in 14 individuals** with severe congenital neutropenia (PMID: 25129144): *"We identify 9 distinct homozygous mutations in the JAGN1 gene encoding Jagunal homolog 1 in 14 individuals with SCN."* The homozygous state across all 14 patients directly supports strict autosomal recessive inheritance.

*JAGN1* encodes **Jagunal homolog 1**, an ER-resident membrane protein of ~189 amino acids with multiple transmembrane domains. The corresponding OMIM entries are phenotype #616022 (SCN7) and gene \*616012. Recurrent and representative pathogenic variants include:

| Variant (cDNA) | Protein    | Class      | Notable association                                                               |
|----------------|------------|------------|-----------------------------------------------------------------------------------|
| c.3G>A         | p.Met1?    | Start-loss | "Pure" neutropenia; rarely extramedullary (except short stature in 1 patient)     |
| c.63G>T        | p.Glu21Asp | Missense   | Syndromic facial features + bone metabolism disorders; found in Romanian patients |
| c.130C>T       | p.His44Tyr | Missense   | Syndromic facial features + bone metabolism disorders                             |
| c.40G>A        | p.Gly14Ser | Missense   | Reported SCN7 allele                                                              |

*JAGN1* deficiency is estimated to account for ~10% of all SCN cases (PMID: 41751872): *"Jagunal homolog 1 (JAGN1) deficiency has been described as a genetic cause of SCN and is now estimated to account for approximately 10% of all SCN cases."*

**Ontology suggestions:** Gene — HGNC:26926 (*JAGN1*); Disease — MONDO:0014865; variant classes — SO:0001583 (missense\_variant), SO:0002012 (start\_lost).

## Finding 2 — *JAGN1* deficiency disrupts the ER secretory pathway, N-glycosylation, and G-CSFR signaling, causing arrested neutrophil maturation and apoptosis

*JAGN1*-mutant granulocytes are "characterized by ultrastructural defects, a paucity of granules, aberrant N-glycosylation of multiple proteins and increased incidence of apoptosis" (PMID: 25129144), and *JAGN1* is required for **G-CSF receptor-mediated signaling**. This establishes the core cellular lesion: a defective early secretory pathway that impairs both the glycoprotein-processing machinery and the surface signaling apparatus of developing neutrophils.

A distinct cell-death program underlies the apoptosis. In myeloid cells expressing SCN-associated *JAGN1* mutants, cells were "remarkably susceptible to agonists that normally trigger degranulation and succumbed to a calcium-dependent cell death programme. This mode of cell death was completely prevented by pharmacological inhibition of calpain but unaffected by caspase inhibition" (PMID: 33206996). Thus the death is **calcium- and calpain-dependent (caspase-independent)** — a mechanistically specific and potentially druggable node.

An in vivo zebrafish model refined the causal ordering, showing that "*Jagn1b* has a critical role in granulocyte colony-stimulating factor receptor signaling and steady-state granulopoiesis, shedding light on the pathogenesis of neutropenia associated with *JAGN1* mutations" (PMID: 38739706), with unfolded protein response (UPR) activation and apoptosis appearing **secondary** to the primary G-CSFR/granulopoiesis defect. Newer work identifies that *Jagn1* preserves **protein sialylation** during neutrophil differentiation, and that its expression is transcriptionally controlled by **C/EBP- $\alpha$**  (PMID: 42685351).

**Ontology suggestions:** GO:0006888 (ER to Golgi vesicle-mediated transport); GO:0006487 (protein N-linked glycosylation); GO:0009101 (glycoprotein biosynthetic process); GO:0006915 (apoptotic process); GO:0030593 (neutrophil chemotaxis); GO:0038158 (granulocyte colony-stimulating factor signaling pathway).

### Finding 3 — Clinical phenotype: neonatal-onset severe neutropenia with recurrent infections, plus a syndromic spectrum

In the ESID/EBMT multicenter cohort of **32 JAGN1-deficient patients**, *"Autosomal recessively inherited variants in JAGN1 lead to congenital neutropenia, early-onset bacterial infections, aphthosis, and skin abscesses due to aberrant differentiation and maturation of neutrophils. Bone metabolism disorders and syndromic phenotype, including facial features, short stature, and neurodevelopmental delay, have been reported"* (PMID: 39775668).

**Core features (essentially all patients):** congenital / early-onset severe neutropenia (ANC  $<0.5 \times 10^9/L$ ), early-onset bacterial infections, aphthous stomatitis, and skin abscesses. In this cohort **all patients** experienced infectious complications.

**Syndromic features (~a third of patients):** In the 32-patient cohort, **12/32 (~38%)** had short stature and facial features, and neurodevelopmental delay occurred in 4 patients from 3 families. Additional reported features include **delayed umbilical cord separation, failure to thrive, bone metabolism disorders (e.g., osteopenia), and recurrent fungal infections** (PMID: 37120535): *"Recurrent abscess formation unresponsive to antibiotic therapy, a history of delayed umbilical separation, frequent bacterial or fungal infection, dysmorphic face, failure to thrive, and other coexisting organ abnormalities should prompt physicians to syndromic immunodeficiencies involving neutrophils."*

**Genotype–phenotype correlation** (PMID: 39775668): *"Variant c.3G>A p.Met1, found in 9 patients, was never connected to extramedullary symptoms, except for short stature in 1 patient. Patients with the variants c.63G>T, p.Glu21Asp and c130c>T p.His44 Tyr presented more often with syndromic facial features and bone metabolism disorders."* In short, the start-loss allele tends toward **"pure" neutropenia**, whereas certain missense alleles produce the **syndromic** end of the spectrum.

| Phenotype                                        | Type             | Onset               | Frequency     | Suggested HPO           |
|--------------------------------------------------|------------------|---------------------|---------------|-------------------------|
| Severe neutropenia (ANC <0.5×10 <sup>9</sup> /L) | Lab abnormality  | Congenital/neonatal | ~all          | HP:0000778 / HP:0001875 |
| Recurrent bacterial infections                   | Clinical         | Infancy             | ~all          | HP:0002718              |
| Recurrent fungal infections                      | Clinical         | Infancy/childhood   | Subset        | HP:0009110              |
| Aphthous stomatitis                              | Clinical sign    | Childhood           | Common        | HP:0011107              |
| Skin abscesses                                   | Clinical sign    | Infancy/childhood   | Common        | HP:0025503 / HP:0100658 |
| Short stature                                    | Physical         | Childhood           | ~38%          | HP:0004322              |
| Facial dysmorphism                               | Physical         | Congenital          | ~38%          | HP:0001999              |
| Neurodevelopmental delay                         | Behavioral/neuro | Childhood           | Subset (4/32) | HP:0012758              |
| Osteopenia / bone metabolism disorder            | Lab/imaging      | Childhood           | Subset        | HP:0000938              |
| Delayed umbilical cord separation                | Clinical sign    | Neonatal            | Subset        | HP:0005249              |
| Failure to thrive                                | Clinical         | Infancy             | Subset        | HP:0001508              |

**Quality of life:** Not formally quantified with instruments (EQ-5D, SF-36) for this ultra-rare disease; morbidity is driven by recurrent infections, hospitalizations, and — where syndromic — growth and developmental impact.

#### Finding 4 — Poor G-CSF response; alloHSCT is the definitive curative treatment

A therapeutically decisive feature of SCN7 is that *"most have been described as low-responders to recombinant granulocyte colony-stimulating factor (G-CSF) therapy"* (PMID: 39286252). In two G-CSF-refractory patients, GM-CSF (sargramostim) was trialed as an alternative but *"GM-CSF did not increase neutrophil counts in our patients"* — and both ultimately underwent successful HSCT.

The multicenter cohort positions transplant clearly (PMID: 39775668): *"Allogeneic hematopoietic stem cell transplantation (alloHSCT) is a treatment option for patients who respond poorly to therapy with G-CSF and those who suffer from complicated infections."* Six of the 32 patients underwent alloHSCT.

Broader SCN evidence frames the trade-offs: long-term G-CSF reduces sepsis mortality but reveals an **MDS/AML predisposition** that plateaus at ~2.3%/year after 10 years (PMID: 20456363), and *"Allogeneic hematopoietic stem cell transplantation (HSCT) is the only curative treatment for SCN"* (PMID: 21072829).

**Ontology suggestions:** NCIT:C1683 (Filgrastim/G-CSF), NCIT:C1863 (Sargramostim/GM-CSF), NCIT:C15431 (Hematopoietic Stem Cell Transplantation), NCIT:C4711 (Allogeneic).

### Finding 5 — JAGN1 is an evolutionarily conserved ER integral-membrane protein that reorganizes the ER and facilitates vesicular/secretory trafficking

The ancestral cell-biological function of JAGN1 was defined in *Drosophila* oogenesis, where Jagunal concentrates the ER into subcortical clusters and is required for vesicular traffic (PMID: 17389229): *"This ER reorganization requires Jagunal, which is an evolutionarily conserved ER membrane protein. Loss of Jagunal reduces vesicular traffic to the oocyte lateral membrane."* Jagn also mediates asymmetric ER partitioning during mitosis in proneural cells (PMID: 28381427), and a genetic-modifier screen links Jagn to Sec63, Presenilin/ $\gamma$ -secretase, and Notch signaling (PMID: 36932646).

Structural/topological reasoning suggests JAGN1 acts in **cargo sorting and vesicle formation**, sharing topology with tetraspanins and Erv cargo-sorting proteins (PMID: 32783652): *"JAGN1, tetraspanins, and Erv proteins: is common topology indicative of common function in cargo sorting?"* This deep conservation explains why loss-of-function produces a defect in the early secretory pathway across species.

**Ontology suggestions:** GO:0005789 (endoplasmic reticulum membrane); GO:0005783 (endoplasmic reticulum); GO:0006888 (ER to Golgi vesicle-mediated transport).

### Finding 6 — Animal and cellular models recapitulate JAGN1-SCN and reveal glycosylation-dependent neutrophil dysfunction rescuable by GM-CSF

A hematopoietic-lineage-specific *Jagn1* knockout mouse demonstrated the qualitative branch of disease. *"Global glycobioime analysis identified marked alterations in the glycosylation of proteins involved in cell adhesion and cytotoxicity in Jagn1-deficient neutrophils"* (PMID: 32783652).

25129145), with "impaired formation of cytotoxic granules, as well as defective myeloperoxidase release and killing of *Candida albicans*." Critically, "GM-CSF also restored the defective fungicidal activity of bone marrow cells from humans with JAGN1 mutations" — a candidate therapeutic mechanism (albeit not borne out for correcting neutrophil counts in the later clinical report,

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P 39286252

The zebrafish *jagn1a/jagn1b* model attributes neutropenia primarily to impaired G-CSFR signaling and steady-state granulopoiesis, with secondary UPR/apoptosis (PMID: 38739706). Cellular models — patient-mutant HL-60 promyelocytes (PMID: 33206996) and Hoxb8-ER-immortalized progenitors (PMID: 42685351) — respectively demonstrate calpain-dependent death and C/EBP- $\alpha$ -controlled, sialylation-dependent adhesion defects (impaired E-selectin and CXCL8 binding).

### Finding 7 — Diagnosis by CBC + bone marrow + NGS; autosomal recessive inheritance with consanguinity/founder contribution

Diagnostic workup rests on **persistent severe neutropenia** (ANC  $<0.5 \times 10^9/L$ ) on repeated CBC with differential, a bone marrow aspirate showing **maturation arrest at the promyelocyte/myelocyte stage** (a general SCN feature), and molecular confirmation. Because JAGN1 is not covered by first-line ELANE/G6PC3 Sanger testing, it requires next-generation sequencing (gene panels or WES). In the Israeli congenital-neutropenia registry, "Sanger sequencing was performed for ELANE or G6PC3, and patients with wild-type ELANE/G6PC3 were referred for next-generation sequencing. Sixty-five patients with neutropenia were included" (PMID: 38600884).

Inheritance is strictly autosomal recessive, and cohorts are enriched for consanguinity: "The Israeli population is characterized by an ethnically diverse population with a high rate of consanguinity" (PMID: 38600884). Recurrent variants show founder/geographic clustering (e.g., c.3G>A p.Met1? across multiple families; c.63G>T p.Glu21Asp including Romanian patients). Homozygosity across all 14 originally described patients (PMID: 25129144) further supports autosomal recessive inheritance with a consanguinity contribution.

### Finding 8 — Prognosis: lifelong disease with infection- and leukemia-related morbidity; prevention centers on prophylaxis, surveillance, and counseling

Untreated SCN carries high early-life mortality from bacterial sepsis. With long-term G-CSF, SCN sepsis mortality falls substantially, but a premalignant MDS/AML predisposition emerges — "Long-term, the annual risk of MDS/AML attained a plateau (2.3%/year after 10 years)" (PMID: 38600884).

20456363). Poor G-CSF responders — the group that includes most JAGN1 patients — fare worst: *"In these less-responsive patients, the cumulative incidence of adverse events was highest: after 10 years, 40% developed MDS/AML and 14% died of sepsis"* (PMID: 16497969).

Because *"SCN is also a premalignant condition; a significant proportion of patients develop myelodysplastic syndrome or leukemia (MDS/L). Allogeneic hematopoietic stem cell transplantation (HSCT) is the only curative treatment for SCN"* (PMID: 21072829), management emphasizes infection prophylaxis, regular hematologic surveillance (CBC, bone marrow with cytogenetics for MDS/AML transformation, monitoring for CSF3R mutations in SCN generally), and genetic counseling for at-risk families. In a congenital-neutropenia registry, ~12% of patients showed myeloid transformation and ~29% required HSCT (PMID: 38600884).

## Finding 9 — Integrated two-branch mechanistic model of SCN7

Convergent evidence across human patient cells, mouse (PMID: 25129145), zebrafish (PMID: 38739706), *Drosophila* (PMID: 17389229), and in-vitro systems (PMID: 33206996, PMID: 42685351) supports a unified model in which a single conserved ER-secretory/glycosylation lesion produces **both** quantitative and qualitative neutrophil defects. The zebrafish model refines earlier views by showing UPR/apoptosis are secondary to the G-CSFR/granulopoiesis defect.

## Mechanistic Model / Interpretation

### Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function JAGN1 variants** (start-loss, missense, etc.) → **loss/reduction of functional Jagunal homolog 1** in the ER membrane of myeloid cells. (*demonstrated — homozygous mutations,*

P 25129144

2. Loss of JAGN1 → **disrupted ER organization and ER-to-Golgi vesicular trafficking / cargo sorting.** (*demonstrated across species;*

P 17389229

P 32783652

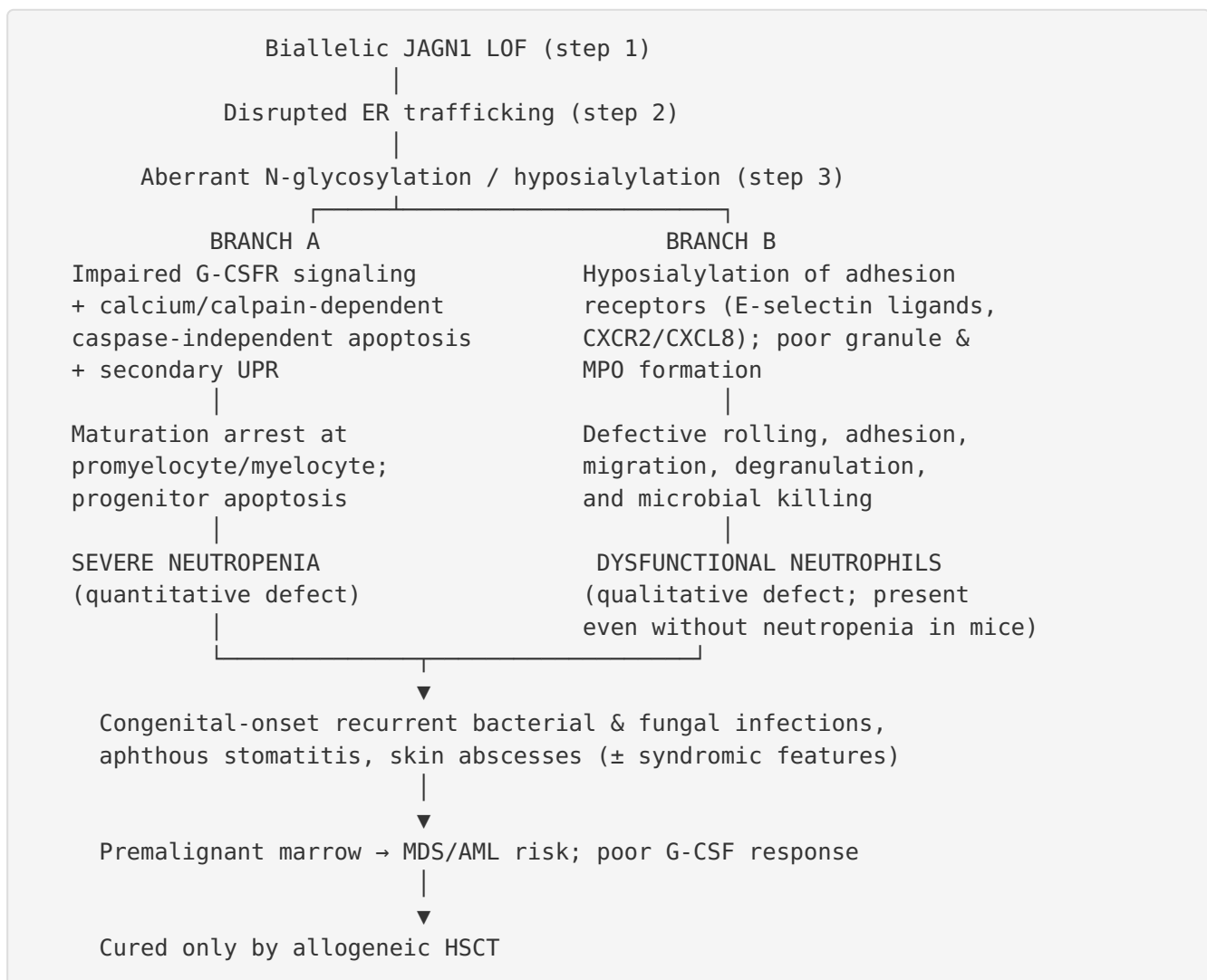
3. Impaired secretory trafficking → **aberrant N-glycosylation and reduced sialylation of glycoproteins**, controlled during differentiation by C/EBP- $\alpha$ . (*demonstrated*;

P 25129144

P 25129145

P 42685351

4. From step 3 the mechanism **branches** (see diagram).



**Branch A (quantitative → neutropenia):** JAGN1 is required for G-CSFR signaling; its loss impairs steady-state granulopoiesis (zebrafish,

P 38739706

and sensitizes myeloid progenitors to a **calcium- and calpain-dependent, caspase-independent** death program (P 33206996),

with UPR/apoptosis as secondary events. The net result is maturation arrest and severe neutropenia.

**Branch B (qualitative → dysfunctional neutrophils):** Even neutrophils that are produced are defective. Hyposialylation of adhesion-relevant receptors impairs E-selectin-dependent rolling and CXCL8/CXCR2-mediated adhesion; cytotoxic granule formation, myeloperoxidase release, and *Candida* killing are all impaired (P 25129145).

(P 42685351)

This branch is demonstrable in mouse models independent of neutrophil count, explaining why infection susceptibility can exceed what the ANC alone would predict.

**Upstream vs downstream:** The mutation and ER-trafficking defect are upstream; glycosylation/sialylation changes are central mediators; G-CSFR signaling failure, apoptosis, and adhesion/killing defects are downstream effectors; infection, marrow premalignancy, and syndromic features are the clinical endpoints.

**Cell types / compartments involved:** neutrophils and their progenitors (promyelocytes, myelocytes) — CL:0000775 (neutrophil), CL:0000557 (granulocyte monocyte progenitor); subcellular compartments — GO:0005789 (ER membrane), GO:0005794 (Golgi apparatus), GO:0042582 (azurophil granule). Anatomy — UBERON:0002371 (bone marrow), UBERON:0000178 (blood), UBERON:0001723 (tongue/oral mucosa for aphthae), UBERON:0002097 (skin).

## Evidence Base

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| PMID                     | Title (abbrev.)                                                                                | Evidence type                   | Supports                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------|
| <a href="#">25129144</a> | <i>JAGN1</i> deficiency causes aberrant myeloid cell homeostasis and congenital neutropenia    | Human genetics/cell             | Causal gene, AR inheritance, core cellular defects (F1, F2, F7)             |
| <a href="#">25129145</a> | <i>Jagunal</i> homolog 1 is a critical regulator of neutrophil function in fungal host defense | Mouse + human cells             | Glycosylation defect, functional killing defect, GM-CSF rescue (F6, F9)     |
| <a href="#">33206996</a> | <i>SCN</i> -associated <i>JAGN1</i> mutations unleash a calpain-dependent cell death programme | In vitro (HL-60)                | Calpain-dependent, caspase-independent apoptosis (F2, F6)                   |
| <a href="#">38739706</a> | <i>JAGN1</i> -associated <i>SCN</i> zebrafish model: altered <i>G-CSFR</i> signaling and UPR   | Zebrafish                       | <i>G-CSFR</i> /granulopoiesis as primary driver; UPR secondary (F2, F6, F9) |
| <a href="#">42685351</a> | <i>Jagn1</i> safeguards neutrophil function by preserving sialylation                          | Mouse/ <i>Hoxb8</i> progenitors | Sialylation, C/EBP- $\alpha$ control, adhesion defect (F2, F6, F9)          |
| <a href="#">39775668</a> | Extended phenotypes/treatment in 32 <i>JAGN1</i> -deficient patients (ESID/EBMT)               | Human cohort (n=32)             | Phenotype spectrum, genotype-phenotype, HSCT (F3, F4)                       |
| <a href="#">37120535</a> | <i>JAGN1</i> mutation with distinct clinical features                                          | Case reports                    | Additional syndromic features (F3)                                          |
| <a href="#">39286252</a> | GM-CSF sargramostim did not rescue neutrophil phenotype in two <i>JAGN1</i> patients           | Case reports                    | Poor G-CSF/GM-CSF response; proceed to HSCT (F4)                            |
| <a href="#">41751872</a> | Phenotypic variability with c.63G>T variant                                                    | Cohort/review                   | ~10% of <i>SCN</i> ; founder clustering (F1, F7)                            |
| <a href="#">38600884</a> | Genetic backgrounds of congenital neutropenias in Israel                                       | Registry                        | NGS diagnostic tiering; consanguinity; transformation/HSCT rates (F7, F8)   |
| <a href="#">17389229</a> |                                                                                                | <i>Drosophila</i>               | Conserved ER-trafficking function (F5, F9)                                  |

| PMID                     | Title (abbrev.)                                                                | Evidence type             | Supports                                                     |
|--------------------------|--------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------|
|                          | <i>Jagunal is required for reorganizing the ER during Drosophila oogenesis</i> |                           |                                                              |
| <a href="#">32783652</a> | <i>JAGN1, tetraspanins, and Erv proteins: common topology/ cargo sorting?</i>  | Computational/ structural | Cargo-sorting hypothesis (F5)                                |
| <a href="#">28381427</a> | <i>ER partitioned asymmetrically during mitosis in proneural cells</i>         | Drosophila                | Jagn in ER partitioning (F5)                                 |
| <a href="#">36932646</a> | <i>Deficiency screen for modifiers of Drosophila Jagunal</i>                   | Drosophila                | Sec63/Presenilin/Notch genetic interactions (F5)             |
| <a href="#">20456363</a> | <i>Stable long-term risk of leukaemia in SCN on G-CSF</i>                      | Registry                  | MDS/AML plateau ~2.3%/yr (F4, F8)                            |
| <a href="#">16497969</a> | <i>Incidence of leukemia and mortality from sepsis in SCN on G-CSF</i>         | Registry                  | Poor responders: 40% MDS/ AML, 14% sepsis death at 10y (F8)  |
| <a href="#">21072829</a> | <i>HSCT in severe congenital neutropenia</i>                                   | Review                    | HSCT the only curative therapy; premalignant nature (F4, F8) |

## Comparative context (SCN subtypes)

| Subtype            | Gene         | Inheritance | G-CSF response     | Distinguishing note                                                                            |
|--------------------|--------------|-------------|--------------------|------------------------------------------------------------------------------------------------|
| SCN1               | <i>ELANE</i> | AD          | Usually responsive | Most common; maturation arrest<br>( <a href="#">P 25427142</a><br><a href="#">P 41037715</a> ) |
| SCN3<br>(Kostmann) | <i>HAX1</i>  | AR          | Responsive         | Neurological features in some transcripts<br>( <a href="#">P 37193639</a> )                    |
| SCN4               | <i>G6PC3</i> | AR          | Variable           | Cardiac/urogenital features, prominent veins<br>( <a href="#">P 32623377</a> )                 |
| SCN7               | <i>JAGN1</i> | AR          | Poor (most)        | ER/glycosylation defect; ~10% of SCN; syndromic in ~1/3                                        |

## Limitations and Knowledge Gaps

- **Small evidence base / ultra-rare disease.** The largest dedicated JAGN1 cohort is 32 patients  
([P 39775668](#)); most other clinical data are case reports. Frequencies (e.g., ~38% with syndromic features) rest on small denominators and are subject to ascertainment bias.
- **No formal epidemiology for SCN7 specifically.** Overall SCN incidence is often cited at ~1–2 per million; the JAGN1 share (~10%) is an estimate. True prevalence, incidence, carrier frequency (gnomAD-derived), sex ratio, and geographic-variant distributions are not precisely established.
- **No validated quality-of-life data** (EQ-5D/SF-36/PROMIS) for SCN7.

- **Mechanistic ordering not fully settled.** Whether UPR/apoptosis is strictly secondary to G-CSFR signaling failure (zebrafish,

( P 38739706 )

versus a parallel driver is not fully resolved in humans; the relative weighting of Branch A vs Branch B for clinical infection risk is inferred, not directly quantified in patients.

- **Therapeutic ambiguity.** GM-CSF rescued fungicidal activity of human cells ex vivo

( P 25129145 )

but failed to raise neutrophil counts in two patients

( P 39286252 );

the discrepancy between functional rescue and count correction is unexplained.

- **Genotype–phenotype correlations are provisional**, based on modest patient numbers; modifier genes and epigenetic contributors in humans are essentially uncharacterized.
- **Animal-model caveats:** zebrafish paralog redundancy (*jagn1a/jagn1b*) and mouse conditional-vs-global differences complicate direct human extrapolation.

## Proposed Follow-up Experiments / Actions

1. **Prospective natural-history registry** for JAGN1-SCN7 through SCNIR/ESID/EBMT to establish incidence, penetrance, sex ratio, MDS/AML transformation rate specific to *JAGN1* (as opposed to pooled SCN), and long-term HSCT outcomes.
2. **Systematic genotype–phenotype mapping** across all reported alleles (c.3G>A start-loss vs c.63G>T/c.130C>T missense) to confirm the "pure vs syndromic" dichotomy with adequate statistical power.
3. **Test calpain inhibition** (e.g., calpeptin/analogous agents) as a mechanism-based strategy to rescue myeloid progenitor survival, building on

( P 33206996 )

4. **Glyco-engineering / sialylation restoration** studies (e.g., supplementation or pathway modulation) in patient-derived iPSC-neutrophils, following the sialylation mechanism

( P 42685351 ).

5. **Resolve the GM-CSF paradox** — dose/route/timing studies and biomarkers to reconcile ex-vivo functional rescue with in-vivo count failure

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P 25129145

vs 39286252).

6. **Gene-correction proof-of-concept.** Given HSCT is curative, adapt CRISPR/HDR autologous HSPC correction approaches (as demonstrated for *ELANE*,

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P 32822592

to *JAGN1*, potentially avoiding allo-transplant morbidity.

7. **Population carrier screening** using gnomAD allele frequencies in consanguineous/founder populations (Israel, Romania) to inform genetic counseling and cascade testing.

8. **Standardized MDS/AML surveillance protocol** (serial marrow cytogenetics, *CSF3R* monitoring) tailored to poor-responder SCN7 patients, who carry the highest transformation risk

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P 16497969

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## Section-by-Section Reference Map

- **1. Disease information / identifiers:** MONDO:0014865; OMIM #616022 (phenotype) / \*616012 (gene); MeSH "Congenital Bone Marrow Failure Syndromes"/"Neutropenia"; ICD-10 D70; Orphanet SCN. Synonyms: JAGN1 deficiency, JAGN1-related severe congenital neutropenia. Derived from aggregated disease-level resources + individual patient reports. (F1)
- **2. Etiology:** Biallelic *JAGN1* LOF (genetic; no environmental/infectious primary cause). Risk factors — consanguinity, founder alleles. No established protective factors. (F1, F7)
- **3. Phenotypes:** See table under Finding 3 with HPO terms.
- **4. Genetic/molecular:** *JAGN1*, HGNC:26926; variants pathogenic/likely pathogenic (ACMG); missense/start-loss; germline; loss-of-function. (F1)
- **5. Environmental:** Not a driver; infections are consequences, not causes. (F3)
- **6. Mechanism:** See causal chain and two-branch model. (F2, F5, F6, F9)
- **7. Anatomy:** Bone marrow (UBERON:0002371), blood, oral mucosa, skin, bone. (F3)

- **8. Temporal:** Congenital/neonatal onset; chronic lifelong; premalignant progression. (F3, F8)
- **9. Inheritance/population:** AR; consanguinity/founder effects; ~10% of SCN. (F1, F7)
- **10. Diagnostics:** CBC (ANC  $<0.5 \times 10^9/L$ ), marrow maturation arrest, NGS/WES. (F7)
- **11. Prognosis:** Infection- and MDS/AML-driven; poor responders worst. (F4, F8)
- **12. Treatment:** G-CSF (often poor response), GM-CSF (failed in reports), supportive antimicrobials, alloHSCT (curative). (F4)
- **13. Prevention:** Infection prophylaxis, MDS/AML surveillance, genetic counseling, prenatal/carrier testing. (F8)
- **14. Other species:** Conserved orthologs — *Drosophila* Jagn, zebrafish *jagn1a/jagn1b*, mouse *Jagn1*; no described natural companion-animal disease. (F5, F6)
- **15. Model organisms:** Mouse (conditional/global KO), zebrafish (morphant/CRISPR/mutant), *Drosophila* (RNAi/deficiency screens), HL-60 and Hoxb8-ER progenitor cell models. Phenotype recapitulation strong for neutropenia and glycosylation/functional defects. (F5, F6)

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*Report compiled from 9 confirmed findings and 30 reviewed papers over 5 investigation iterations. Evidence types span human clinical/genetic cohorts, mouse and zebrafish in-vivo models, in-vitro cell systems, and computational/structural analyses.*

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